

Vehicle Agent - Frameworks Report

pitstrack_api - AI agent for querying fleet/vehicle data

1. Overview

The Vehicle Agent is a conversational AI assistant that answers natural-language questions about a company's vehicle fleet (odometer, speed, status, driver, location, etc.) by pulling live data from the Pitstrack API.

2. Core Frameworks Used

Framework / Library	Role in the agent
LangChain	Agent construction (create_agent) and tool-calling abstraction
LangGraph	Underlying graph engine that runs the agent loop and tool-call routing
langgraph-checkpoint-mysql	Persistent, database-backed conversation memory - see Section 3
Ollama	Runs the local LLM (qwen3:1.7b) that powers the agent
FastAPI	REST API layer exposing the vehicle endpoints that the agent's tools call

3. Memory Improvements

By default, a LangGraph agent's conversation state lives only in memory and is lost when the process restarts. This project replaces that with a custom checkpointer that persists every conversation turn to a MySQL database using langgraph-checkpoint-mysql. Each conversation is identified by a thread_id, so history is durable across restarts and isolated per conversation.